GROTHENDIECK EXISTENCE FOR COHERENT SYSTEMS WITH PROPER SUPPORT ON A SEPARATED FINITE-TYPE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Public376 counts projective algebraization. This new statement allows a separated finite-type ambient X that need not be proper and requires proper support; the projective result is retained as uncounted support.

Editorial note. The earlier proper case and its local lemmas are included for completeness but are not additional counted tasks.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Coherent formal systems, Artin–Rees and formal functions, proper coherent pushforward, Chow lemma, Noetherian induction and support under proper maps. Projective/proper existence and the local push-pull/error and support lemmas are retained author proof, not given away as the target theorem.

Source reference. Original section 0EHN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18138–18138. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original equation 0880: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10445–10445. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01CQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1771–1771. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0882: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10447–10447. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 087U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10439–10439. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Date: stacks-project:coherent.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 087Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10444–10444. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0897: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10487–10487. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087Y: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10443–10443. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10442–10442. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0881: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10446–10446. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10441–10441. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 087T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10438–10438. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0889: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10454–10454. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 02O5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3504–3504. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02OB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3510–3510. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 02OC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3511–3511. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00MB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 817–817. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02KH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3372–3372. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05EC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6985–6985. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Y9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2548–2548. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Y8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2547–2547. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0200: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2617–2617. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01J4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2001–2001. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0C4H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15227–15227. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0C4J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15229–15229. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2468–2468. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09MQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12188–12188. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01T5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2362–2362. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01T3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2360–2360. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03GN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4534–4534. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2361–2361. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01JV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2028–2028. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01W4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2469–2469. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AH6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13219–13219. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 035C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4123–4123. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CYI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16243–16243. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01WN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2488–2488. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 056J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6712–6712. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 05JV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7179–7179. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01B6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1715–1715. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 02MO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3451–3451. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 01LA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2079–2079. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 01B4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1713–1713. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01T6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2363–2363. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 01XY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2537–2537. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01Y6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2545–2545. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01F6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1859–1859. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 02MP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3452–3452. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01WA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2475–2475. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CYJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16244–16244. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7062–7062. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1 (Grothendieck's existence theorem)**.** [1, III Theorem 5.1.5] *Let A be a Noetherian ring complete with respect to an ideal I. Let X be a separated, finite type scheme over A. Then the functor (4.0.1)*

$$Coh_{\text{support proper over } A}(\mathcal{O}_X) \longrightarrow Coh_{\text{support proper over } A}(X, \mathcal{I})$$

is an equivalence.

1. Grothendieck's existence theorem, I

In this section we discuss Grothendieck's existence theorem for the projective case. We will use the notion of coherent formal modules developed in Section 0EHN. The reader who is familiar with formal schemes is encouraged to read the statement and proof of the theorem in [1].

Lemma 1.1. *Let A be Noetherian ring complete with respect to an ideal I. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism. Let $\mathcal{I} = I\mathcal{O}_X$. Then the functor (0880) is fully faithful.*

Proof. Let $\mathcal{F}$, $\mathcal{G}$ be coherent $\mathcal{O}_X$-modules. Then $\mathcal{H} = \mathcal{H}om_{\mathcal{O}_X}(\mathcal{G}, \mathcal{F})$ is a coherent $\mathcal{O}_X$-module, see Modules, Lemma 01CQ. By Lemma 0882 the map

$$\lim_n H^0(X, \mathcal{H}/\mathcal{I}^n\mathcal{H}) \to \mathrm{Mor}_{Coh(X,\mathcal{I})}(\mathcal{G}^\wedge, \mathcal{F}^\wedge)$$

is bijective. Hence fully faithfulness of (0880) follows from the theorem on formal functions (Lemma 087U) for the coherent sheaf $\mathcal{H}$. □

Lemma 1.2. *Let A be Noetherian ring and $I \subset A$ an ideal. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism and let $\mathcal{L}$ be an f-ample invertible sheaf. Let $\mathcal{I} = I\mathcal{O}_X$. Let $(\mathcal{F}_n)$ be an object of $Coh(X, \mathcal{I})$. Then there exists an integer d_0 such that*

$$H^1(X, \mathrm{Ker}(\mathcal{F}_{n+1} \to \mathcal{F}_n) \otimes \mathcal{L}^{\otimes d}) = 0$$

for all $n \geq 0$ and all $d \geq d_0$.

Proof. Set $B = \bigoplus I^n/I^{n+1}$ and $\mathcal{B} = \bigoplus \mathcal{I}^n/\mathcal{I}^{n+1} = f^*\widetilde{B}$. By Lemma 087Z the graded quasi-coherent $\mathcal{B}$-module $\mathcal{G} = \bigoplus \mathrm{Ker}(\mathcal{F}_{n+1} \to \mathcal{F}_n)$ is of finite type. Hence the lemma follows from Lemma 0897 part (2). □

Lemma 1.3. *Let A be Noetherian ring complete with respect to an ideal I. Let $f : X \to \mathrm{Spec}(A)$ be a projective morphism. Let $\mathcal{I} = I\mathcal{O}_X$. Then the functor (0880) is an equivalence.*

Proof. We have already seen that (0880) is fully faithful in Lemma 1.1. Thus it suffices to show that the functor is essentially surjective.

We first show that every object $(\mathcal{F}_n)$ of $Coh(X, \mathcal{I})$ is the quotient of an object in the image of (0880). Let $\mathcal{L}$ be an f-ample invertible sheaf on X. Choose d_0 as in Lemma 1.2. Choose a $d \geq d_0$ such that $\mathcal{F}_1 \otimes \mathcal{L}^{\otimes d}$ is globally generated by some sections $s_{1,1}, \ldots, s_{t,1}$. Since the transition maps of the system

$$H^0(X, \mathcal{F}_{n+1} \otimes \mathcal{L}^{\otimes d}) \longrightarrow H^0(X, \mathcal{F}_n \otimes \mathcal{L}^{\otimes d})$$

are surjective by the vanishing of H^1 we can lift $s_{1,1}, \ldots, s_{t,1}$ to a compatible system of global sections $s_{1,n}, \ldots, s_{t,n}$ of $\mathcal{F}_n \otimes \mathcal{L}^{\otimes d}$. These determine a compatible system of maps

$$(s_{1,n}, \ldots, s_{t,n}) : (\mathcal{L}^{\otimes -d})^{\oplus t} \longrightarrow \mathcal{F}_n$$

Using Lemma 087Y we deduce that we have a surjective map

$$\left((\mathcal{L}^{\otimes -d})^{\oplus t}\right)^\wedge \longrightarrow (\mathcal{F}_n)$$

as desired.

The result of the previous paragraph and the fact that $Coh(X, \mathcal{I})$ is abelian (Lemma 087X) implies that every object of $Coh(X, \mathcal{I})$ is a cokernel of a map between objects coming from $Coh(\mathcal{O}_X)$. As (0880) is fully faithful and exact by Lemmas 1.1 and 0881 we conclude. □

2. Grothendieck's existence theorem, II

In this section we discuss Grothendieck's existence theorem in the proper case. Before we give the statement and proof, we need to develop a bit more theory regarding the categories $Coh(X, \mathcal{I})$ of coherent formal modules introduced in Section 0EHN.

Remark 2.1. Let X be a Noetherian scheme and let $\mathcal{I}, \mathcal{K} \subset \mathcal{O}_X$ be quasi-coherent sheaves of ideals. Let $\alpha : (\mathcal{F}_n) \to (\mathcal{G}_n)$ be a morphism of $Coh(X, \mathcal{I})$. Given an affine open $\mathrm{Spec}(A) = U \subset X$ with $\mathcal{I}|_U, \mathcal{K}|_U$ corresponding to ideals $I, K \subset A$ denote $\alpha_U : M \to N$ of finite $A^\wedge$-modules which corresponds to $\alpha|_U$ via Lemma 087W. We claim the following are equivalent

(1) there exists an integer $t \geq 1$ such that $\mathrm{Ker}(\alpha_n)$ and $\mathrm{Coker}(\alpha_n)$ are annihilated by $\mathcal{K}^t$ for all $n \geq 1$,
(2) for any affine open $\mathrm{Spec}(A) = U \subset X$ as above the modules $\mathrm{Ker}(\alpha_U)$ and $\mathrm{Coker}(\alpha_U)$ are annihilated by K^t for some integer $t \geq 1$, and
(3) there exists a finite affine open covering $X = \bigcup U_i$ such that the conclusion of (2) holds for α_{U_i}.

If these equivalent conditions hold we will say that α is a *map whose kernel and cokernel are annihilated by a power of* $\mathcal{K}$. To see the equivalence we use the following commutative algebra fact: suppose given an exact sequence

$$0 \to T \to M \to N \to Q \to 0$$

of A-modules with T and Q annihilated by K^t for some ideal $K \subset A$. Then for every $f, g \in K^t$ there exists a canonical map "fg" : $N \to M$ such that $M \to N \to M$ is equal to multiplication by fg. Namely, for $y \in N$ we can pick $x \in M$ mapping to fy in N and then we can set "fg"$(y) = gx$. Thus it is clear that $\mathrm{Ker}(M/JM \to N/JN)$ and $\mathrm{Coker}(M/JM \to N/JN)$ are annihilated by K^{2t} for any ideal $J \subset A$.

Applying the commutative algebra fact to α_{U_i} and $J = I^n$ we see that (3) implies (1). Conversely, suppose (1) holds and $M \to N$ is equal to α_U. Then there is a $t \geq 1$ such that $\mathrm{Ker}(M/I^nM \to N/I^nN)$ and $\mathrm{Coker}(M/I^nM \to N/I^nN)$ are annihilated by K^t for all n. We obtain maps "fg" : $N/I^nN \to M/I^nM$ which in the limit induce a map $N \to M$ as N and M are I-adically complete. Since the composition with $N \to M \to N$ is multiplication by fg we conclude that fg annihilates T and Q. In other words T and Q are annihilated by K^{2t} as desired.

Lemma 2.2. *Let X be a Noetherian scheme. Let $\mathcal{I}, \mathcal{K} \subset \mathcal{O}_X$ be quasi-coherent sheaves of ideals. Let $X_e \subset X$ be the closed subscheme cut out by $\mathcal{K}^e$. Let $\mathcal{I}_e = \mathcal{I}\mathcal{O}_{X_e}$. Let $(\mathcal{F}_n)$ be an object of $Coh(X, \mathcal{I})$. Assume*

(1) *the functor $Coh(\mathcal{O}_{X_e}) \to Coh(X_e, \mathcal{I}_e)$ is an equivalence for all $e \geq 1$, and*
(2) *there exists a coherent sheaf $\mathcal{H}$ on X and a map $\alpha : (\mathcal{F}_n) \to \mathcal{H}^\wedge$ whose kernel and cokernel are annihilated by a power of $\mathcal{K}$.*

Then $(\mathcal{F}_n)$ is in the essential image of (0880).

Proof. During this proof we will use without further mention that for a closed immersion $i : Z \to X$ the functor i_* gives an equivalence between the category of coherent modules on Z and coherent modules on X annihilated by the ideal sheaf of Z, see Lemma 087T. In particular we may identify $Coh(\mathcal{O}_{X_e})$ with the category of coherent $\mathcal{O}_X$-modules annihilated by $\mathcal{K}^e$ and $Coh(X_e, \mathcal{I}_e)$ as the full subcategory of $Coh(X, \mathcal{I})$ of objects annihilated by $\mathcal{K}^e$. Moreover (1) tells us these two categories are equivalent under the completion functor (0880).

Applying this equivalence we get a coherent $\mathcal{O}_X$-module $\mathcal{G}_e$ annihilated by $\mathcal{K}^e$ corresponding to the system $(\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n)$ of $Coh(X, \mathcal{I})$. The maps $\mathcal{F}_n/\mathcal{K}^{e+1}\mathcal{F}_n \to$

$\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n$ correspond to canonical maps $\mathcal{G}_{e+1} \to \mathcal{G}_e$ which induce isomorphisms $\mathcal{G}_{e+1}/\mathcal{K}^e\mathcal{G}_{e+1} \to \mathcal{G}_e$. Hence $(\mathcal{G}_e)$ is an object of $Coh(X, \mathcal{K})$. The map α induces a system of maps

$$\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n \longrightarrow \mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H}$$

whence maps $\mathcal{G}_e \to \mathcal{H}/\mathcal{K}^e\mathcal{H}$ (by the equivalence of categories again). Let $t \geq 1$ be an integer, which exists by assumption (2), such that $\mathcal{K}^t$ annihilates the kernel and cokernel of all the maps $\mathcal{F}_n \to \mathcal{H}/\mathcal{I}^n\mathcal{H}$. Then $\mathcal{K}^{2t}$ annihilates the kernel and cokernel of the maps $\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n \to \mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H}$, see Remark 2.1. Whereupon we conclude that $\mathcal{K}^{4t}$ annihilates the kernel and the cokernel of the maps

$$\mathcal{G}_e \longrightarrow \mathcal{H}/\mathcal{K}^e\mathcal{H},$$

see Remark 2.1. We apply Lemma 0889 to obtain a coherent $\mathcal{O}_X$-module $\mathcal{F}$, a map $a : \mathcal{F} \to \mathcal{H}$ and an isomorphism $\beta : (\mathcal{G}_e) \to (\mathcal{F}/\mathcal{K}^e\mathcal{F})$ in $Coh(X, \mathcal{K})$. Working backwards, for a given n the triple $(\mathcal{F}/\mathcal{I}^n\mathcal{F}, a \bmod \mathcal{I}^n, \beta \bmod \mathcal{I}^n)$ is a triple as in the lemma for the morphism $\alpha_n \bmod \mathcal{K}^e : (\mathcal{F}_n/\mathcal{K}^e\mathcal{F}_n) \to (\mathcal{H}/(\mathcal{I}^n + \mathcal{K}^e)\mathcal{H})$ of $Coh(X, \mathcal{K})$. Thus the uniqueness in Lemma 0889 gives a canonical isomorphism $\mathcal{F}/\mathcal{I}^n\mathcal{F} \to \mathcal{F}_n$ compatible with all the morphisms in sight. This finishes the proof of the lemma. □

Lemma 2.3. *Let Y be a Noetherian scheme. Let $\mathcal{J}, \mathcal{K} \subset \mathcal{O}_Y$ be quasi-coherent sheaves of ideals. Let $f : X \to Y$ be a proper morphism which is an isomorphism over $V = Y \setminus V(\mathcal{K})$. Set $\mathcal{I} = f^{-1}\mathcal{J}\mathcal{O}_X$. Let $(\mathcal{G}_n)$ be an object of $Coh(Y, \mathcal{J})$, let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module, and let $\beta : (f^*\mathcal{G}_n) \to \mathcal{F}^\wedge$ be an isomorphism in $Coh(X, \mathcal{I})$. Then there exists a map*

$$\alpha : (\mathcal{G}_n) \longrightarrow (f_*\mathcal{F})^\wedge$$

in $Coh(Y, \mathcal{J})$ whose kernel and cokernel are annihilated by a power of $\mathcal{K}$.

Proof. Since f is a proper morphism we see that $f_*\mathcal{F}$ is a coherent $\mathcal{O}_Y$-module (Proposition 02O5). Thus the statement of the lemma makes sense. Consider the compositions

$$\gamma_n : \mathcal{G}_n \to f_*f^*\mathcal{G}_n \to f_*(\mathcal{F}/\mathcal{I}^n\mathcal{F}).$$

Here the first map is the adjunction map and the second is $f_*\beta_n$. We claim that there exists a unique α as in the lemma such that the compositions

$$\mathcal{G}_n \xrightarrow{\alpha_n} f_*\mathcal{F}/\mathcal{J}^n f_*\mathcal{F} \to f_*(\mathcal{F}/\mathcal{I}^n\mathcal{F})$$

equal γ_n for all n. Because of the uniqueness we may assume that $Y = \mathrm{Spec}(B)$ is affine. Let $J \subset B$ corresponds to the ideal $\mathcal{J}$. Set

$$M_n = H^0(X, \mathcal{F}/\mathcal{I}^n\mathcal{F}) \quad \text{and} \quad M = H^0(X, \mathcal{F})$$

By Lemma 02OB and Theorem 02OC the inverse limit of the modules M_n equals the completion $M^\wedge = \lim M/J^nM$. Set $N_n = H^0(Y, \mathcal{G}_n)$ and $N = \lim N_n$. Via the equivalence of categories of Lemma 087W the finite $B^\wedge$ modules N and $M^\wedge$ correspond to $(\mathcal{G}_n)$ and $f_*\mathcal{F}^\wedge$. It follows from this that α has to be the morphism of $Coh(Y, \mathcal{J})$ corresponding to the homomorphism

$$\lim \gamma_n : N = \lim_n N_n \longrightarrow \lim M_n = M^\wedge$$

of finite $B^\wedge$-modules.

We still have to show that the kernel and cokernel of α are annihilated by a power of $\mathcal{K}$. Set $Y' = \mathrm{Spec}(B^\wedge)$ and $X' = Y' \times_Y X$. Let $\mathcal{K}'$, $\mathcal{J}'$, $\mathcal{G}'_n$ and $\mathcal{I}'$, $\mathcal{F}'$ be the pullback of $\mathcal{K}$, $\mathcal{J}$, $\mathcal{G}_n$ and $\mathcal{I}$, $\mathcal{F}$, to Y' and X'. The projection morphism $f' : X' \to Y'$ is the base change of f by $Y' \to Y$. Note that $Y' \to Y$ is a flat morphism of schemes as $B \to B^\wedge$ is flat by Algebra, Lemma 00MB. Hence $f'_*\mathcal{F}'$, resp. $f'_*(f')^*\mathcal{G}'_n$ is the pullback of $f_*\mathcal{F}$, resp. $f_*f^*\mathcal{G}_n$ to Y' by Lemma 02KH. The uniqueness of our construction shows the pullback of α to Y' is the corresponding

map α' constructed for the situation on Y'. Moreover, to check that the kernel and cokernel of α are annihilated by $\mathcal{K}^t$ it suffices to check that the kernel and cokernel of α' are annihilated by $(\mathcal{K}')^t$. Namely, to see this we need to check this for kernels and cokernels of the maps α_n and α'_n (see Remark 2.1) and the ring map $B \to B^\wedge$ induces an equivalence of categories between modules annihilated by J^n and $(J')^n$, see More on Algebra, Lemma 05EC. Thus we may assume B is complete with respect to J.

Assume $Y = \mathrm{Spec}(B)$ is affine, $\mathcal{J}$ corresponds to the ideal $J \subset B$, and B is complete with respect to J. In this case $(\mathcal{G}_n)$ is in the essential image of the functor $Coh(\mathcal{O}_Y) \to Coh(Y, \mathcal{J})$. Say $\mathcal{G}$ is a coherent $\mathcal{O}_Y$-module such that $(\mathcal{G}_n) = \mathcal{G}^\wedge$. Note that $f^*(\mathcal{G}^\wedge) = (f^*\mathcal{G})^\wedge$. Hence Lemma 1.1 tells us that β comes from an isomorphism $b : f^*\mathcal{G} \to \mathcal{F}$ and α is the completion functor applied to

$$\mathcal{G} \to f_*f^*\mathcal{G} \cong f_*\mathcal{F}$$

Hence we are trying to verify that the kernel and cokernel of the adjunction map $c : \mathcal{G} \to f_*f^*\mathcal{G}$ are annihilated by a power of $\mathcal{K}$. However, since the restriction $f|_{f^{-1}(V)} : f^{-1}(V) \to V$ is an isomorphism we see that $c|_V$ is an isomorphism. Thus the coherent sheaves $\mathrm{Ker}(c)$ and $\mathrm{Coker}(c)$ are supported on $V(\mathcal{K})$ hence are annihilated by a power of $\mathcal{K}$ (Lemma 01Y9) as desired. □

The following proposition is the form of Grothendieck's existence theorem which is most often used in practice.

Proposition 2.4. *Let A be a Noetherian ring complete with respect to an ideal I. Let $f : X \to \mathrm{Spec}(A)$ be a proper morphism of schemes. Set $\mathcal{I} = I\mathcal{O}_X$. Then the functor (0880) is an equivalence.*

Proof. We have already seen that (0880) is fully faithful in Lemma 1.1. Thus it suffices to show that the functor is essentially surjective.

Consider the collection Ξ of quasi-coherent sheaves of ideals $\mathcal{K} \subset \mathcal{O}_X$ such that every object $(\mathcal{F}_n)$ annihilated by $\mathcal{K}$ is in the essential image. We want to show (0) is in Ξ. If not, then since X is Noetherian there exists a maximal quasi-coherent sheaf of ideals $\mathcal{K}$ not in Ξ, see Lemma 01Y8. After replacing X by the closed subscheme of X corresponding to $\mathcal{K}$ we may assume that every nonzero $\mathcal{K}$ is in Ξ. (This uses the correspondence by coherent modules annihilated by $\mathcal{K}$ and coherent modules on the closed subscheme corresponding to $\mathcal{K}$, see Lemma 087T.) Let $(\mathcal{F}_n)$ be an object of $Coh(X, \mathcal{I})$. We will show that this object is in the essential image of the functor (0880), thereby completion the proof of the proposition.

Apply Chow's lemma (Lemma 0200) to find a proper surjective morphism $f : X' \to X$ which is an isomorphism over a dense open $U \subset X$ such that X' is projective over A. Let $\mathcal{K}$ be the quasi-coherent sheaf of ideals cutting out the reduced complement $X \setminus U$. By the projective case of Grothendieck's existence theorem (Lemma 1.3) there exists a coherent module $\mathcal{F}'$ on X' such that $(\mathcal{F}')^\wedge \cong (f^*\mathcal{F}_n)$. By Proposition 02O5 the $\mathcal{O}_X$-module $\mathcal{H} = f_*\mathcal{F}'$ is coherent and by Lemma 2.3 there exists a morphism $(\mathcal{F}_n) \to \mathcal{H}^\wedge$ of $Coh(X, \mathcal{I})$ whose kernel and cokernel are annihilated by a power of $\mathcal{K}$. The powers $\mathcal{K}^e$ are all in Ξ so that (0880) is an equivalence for the closed subschemes $X_e = V(\mathcal{K}^e)$. We conclude by Lemma 2.2. □

3. Being proper over a base

This is just a short section to point out some useful features of closed subsets proper over a base and finite type, quasi-coherent modules with support proper over a base.

Lemma 3.1. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $Z \subset X$ be a closed subset. The following are equivalent*

(1) *the morphism $Z \to S$ is proper if Z is endowed with the reduced induced closed subscheme structure (Schemes, Definition 01J4),*
(2) *for some closed subscheme structure on Z the morphism $Z \to S$ is proper,*
(3) *for any closed subscheme structure on Z the morphism $Z \to S$ is proper.*

Proof. The implications (3) $\Rightarrow$ (1) and (1) $\Rightarrow$ (2) are immediate. Thus it suffices to prove that (2) implies (3). We urge the reader to find their own proof of this fact. Let Z' and Z'' be closed subscheme structures on Z such that $Z' \to S$ is proper. We have to show that $Z'' \to S$ is proper. Let $Z''' = Z' \cup Z''$ be the scheme theoretic union, see Morphisms, Definition 0C4H. Then Z''' is another closed subscheme structure on Z. This follows for example from the description of scheme theoretic unions in Morphisms, Lemma 0C4J. Since $Z'' \to Z'''$ is a closed immersion it suffices to prove that $Z''' \to S$ is proper (see Morphisms, Lemmas 01W5 and 01W3). The morphism $Z' \to Z'''$ is a bijective closed immersion and in particular surjective and universally closed. Then the fact that $Z' \to S$ is separated implies that $Z''' \to S$ is separated, see Morphisms, Lemma 09MQ. Moreover $Z''' \to S$ is locally of finite type as $X \to S$ is locally of finite type (Morphisms, Lemmas 01T5 and 01T3). Since $Z' \to S$ is quasi-compact and $Z' \to Z'''$ is a homeomorphism we see that $Z''' \to S$ is quasi-compact. Finally, since $Z' \to S$ is universally closed, we see that the same thing is true for $Z''' \to S$ by Morphisms, Lemma 03GN. This finishes the proof. □

Definition 3.2. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $Z \subset X$ be a closed subset. We say *Z is proper over S* if the equivalent conditions of Lemma 3.1 are satisfied.

The lemma used in the definition above is false if the morphism $f : X \to S$ is not locally of finite type. Therefore we urge the reader not to use this terminology if f is not locally of finite type.

Lemma 3.3. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $Y \subset Z \subset X$ be closed subsets. If Z is proper over S, then the same is true for Y.*

Proof. Omitted. □

Lemma 3.4. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with f locally of finite type. If Z is a closed subset of X proper over S, then $(g')^{-1}(Z)$ is a closed subset of X' proper over S'.

Proof. Observe that the statement makes sense as f' is locally of finite type by Morphisms, Lemma 01T4. Endow Z with the reduced induced closed subscheme structure. Denote $Z' = (g')^{-1}(Z)$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z' = X' \times_X Z = (S' \times_S X) \times_X Z = S' \times_S Z$ is proper over S' as a base change of Z over S (Morphisms, Lemma 01W4). □

Lemma 3.5. *Let S be a scheme. Let $f : X \to Y$ be a morphism of schemes which are locally of finite type over S.*

(1) *If Y is separated over S and $Z \subset X$ is a closed subset proper over S, then $f(Z)$ is a closed subset of Y proper over S.*
(2) *If f is universally closed and $Z \subset X$ is a closed subset proper over S, then $f(Z)$ is a closed subset of Y proper over S.*

(3) *If f is proper and $Z \subset Y$ is a closed subset proper over S, then $f^{-1}(Z)$ is a closed subset of X proper over S.*

Proof. Proof of (1). Assume Y is separated over S and $Z \subset X$ is a closed subset proper over S. Endow Z with the reduced induced closed subscheme structure and apply Morphisms, Lemma 0AH6 to $Z \to Y$ over S to conclude.

Proof of (2). Assume f is universally closed and $Z \subset X$ is a closed subset proper over S. Endow Z and $Z' = f(Z)$ with their reduced induced closed subscheme structures. We obtain an induced morphism $Z \to Z'$. Denote $Z'' = f^{-1}(Z')$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z'' \to Z'$ is universally closed as a base change of f (Morphisms, Lemma 01W4). Hence $Z \to Z'$ is universally closed as a composition of the closed immersion $Z \to Z''$ and $Z'' \to Z'$ (Morphisms, Lemmas 01W5 and 01W3). We conclude that $Z' \to S$ is separated by Morphisms, Lemma 09MQ. Since $Z \to S$ is quasi-compact and $Z \to Z'$ is surjective we see that $Z' \to S$ is quasi-compact. Since $Z' \to S$ is the composition of $Z' \to Y$ and $Y \to S$ we see that $Z' \to S$ is locally of finite type (Morphisms, Lemmas 01T5 and 01T3). Finally, since $Z \to S$ is universally closed, we see that the same thing is true for $Z' \to S$ by Morphisms, Lemma 03GN. This finishes the proof.

Proof of (3). Assume f is proper and $Z \subset Y$ is a closed subset proper over S. Endow Z with the reduced induced closed subscheme structure. Denote $Z' = f^{-1}(Z)$ the scheme theoretic inverse image (Schemes, Definition 01JV). Then $Z' \to Z$ is proper as a base change of f (Morphisms, Lemma 01W4). Whence $Z' \to S$ is proper as the composition of $Z' \to Z$ and $Z \to S$ (Morphisms, Lemma 01W3). This finishes the proof. □

Lemma 3.6. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $Z_i \subset X$, $i = 1, \ldots, n$ be closed subsets. If Z_i, $i = 1, \ldots, n$ are proper over S, then the same is true for $Z_1 \cup \ldots \cup Z_n$.*

Proof. Endow Z_i with their reduced induced closed subscheme structures. The morphism

$$Z_1 \amalg \ldots \amalg Z_n \longrightarrow X$$

is finite by Morphisms, Lemmas 035C and 0CYI. As finite morphisms are universally closed (Morphisms, Lemma 01WN) and since $Z_1 \amalg \ldots \amalg Z_n$ is proper over S we conclude by Lemma 3.5 part (2) that the image $Z_1 \cup \ldots \cup Z_n$ is proper over S. □

Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. Then the support $\mathrm{Supp}(\mathcal{F})$ of $\mathcal{F}$ is a closed subset of X, see Morphisms, Lemma 056J. Hence it makes sense to say "the support of $\mathcal{F}$ is proper over S".

Lemma 3.7. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. The following are equivalent*

(1) *the support of $\mathcal{F}$ is proper over S,*
(2) *the scheme theoretic support of $\mathcal{F}$ (Morphisms, Definition 05JV) is proper over S, and*
(3) *there exists a closed subscheme $Z \subset X$ and a finite type, quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ such that (a) $Z \to S$ is proper, and (b) $(Z \to X)_*\mathcal{G} = \mathcal{F}$.*

Proof. The support $\mathrm{Supp}(\mathcal{F})$ of $\mathcal{F}$ is a closed subset of X, see Morphisms, Lemma 056J. Hence we can apply Definition 3.2. Since the scheme theoretic support of $\mathcal{F}$ is a closed subscheme whose underlying closed subset is $\mathrm{Supp}(\mathcal{F})$ we see that (1)

and (2) are equivalent by Definition 3.2. It is clear that (2) implies (3). Conversely, if (3) is true, then $\mathrm{Supp}(\mathcal{F}) \subset Z$ (an inclusion of closed subsets of X) and hence $\mathrm{Supp}(\mathcal{F})$ is proper over S for example by Lemma 3.3. □

Lemma 3.8. *Consider a cartesian diagram of schemes*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

with f locally of finite type. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. If the support of $\mathcal{F}$ is proper over S, then the support of $(g')^*\mathcal{F}$ is proper over S'.

Proof. Observe that the statement makes sense because $(g') * \mathcal{F}$ is of finite type by Modules, Lemma 01B6. We have $\mathrm{Supp}((g')^*\mathcal{F}) = (g')^{-1}(\mathrm{Supp}(\mathcal{F}))$ by Morphisms, Lemma 056J. Thus the lemma follows from Lemma 3.4. □

Lemma 3.9. *Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$, $\mathcal{G}$ be finite type, quasi-coherent $\mathcal{O}_X$-module.*

1. *If the supports of $\mathcal{F}$, $\mathcal{G}$ are proper over S, then the same is true for $\mathcal{F} \oplus \mathcal{G}$, for any extension of $\mathcal{G}$ by $\mathcal{F}$, for $\mathrm{Im}(u)$ and $\mathrm{Coker}(u)$ given any $\mathcal{O}_X$-module map $u : \mathcal{F} \to \mathcal{G}$, and for any quasi-coherent quotient of $\mathcal{F}$ or $\mathcal{G}$.*
2. *If S is locally Noetherian, then the category of coherent $\mathcal{O}_X$-modules with support proper over S is a Serre subcategory (Homology, Definition 02MO) of the abelian category of coherent $\mathcal{O}_X$-modules.*

Proof. Proof of (1). Let Z, Z' be the support of $\mathcal{F}$ and $\mathcal{G}$. Then all the sheaves mentioned in (1) have support contained in $Z \cup Z'$. Thus the assertion itself is clear from Lemmas 3.3 and 3.6 provided we check that these sheaves are finite type and quasi-coherent. For quasi-coherence we refer the reader to Schemes, Section 01LA. For "finite type" we suggest the reader take a look at Modules, Section 01B4.

Proof of (2). The proof is the same as the proof of (1). Note that the assertions make sense as X is locally Noetherian by Morphisms, Lemma 01T6 and by the description of the category of coherent modules in Section 01XY. □

Lemma 3.10. *Let S be a locally Noetherian scheme. Let $f : X \to S$ be a morphism of schemes which is locally of finite type. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module with support proper over S. Then $R^p f_*\mathcal{F}$ is a coherent $\mathcal{O}_S$-module for all $p \geq 0$.*

Proof. By Lemma 3.7 there exists a closed immersion $i : Z \to X$ and a finite type, quasi-coherent $\mathcal{O}_Z$-module $\mathcal{G}$ such that (a) $g = f \circ i : Z \to S$ is proper, and (b) $i_*\mathcal{G} = \mathcal{F}$. We see that $R^p g_*\mathcal{G}$ is coherent on S by Proposition 02O5. On the other hand, $R^q i_*\mathcal{G} = 0$ for $q > 0$ (Lemma 01Y6). By Cohomology, Lemma 01F6 we get $R^p f_*\mathcal{F} = R^p g_*\mathcal{G}$ which concludes the proof. □

Lemma 3.11. *Let S be a Noetherian scheme. Let $f : X \to S$ be a finite type morphism. Let $\mathcal{I} \subset \mathcal{O}_X$ be a quasi-coherent sheaf of ideals. The following are Serre subcategories of $Coh(X, \mathcal{I})$*

1. *the full subcategory of $Coh(X, \mathcal{I})$ consisting of those objects $(\mathcal{F}_n)$ such that the support of $\mathcal{F}_1$ is proper over S,*
2. *the full subcategory of $Coh(X, \mathcal{I})$ consisting of those objects $(\mathcal{F}_n)$ such that there exists a closed subscheme $Z \subset X$ proper over S with $\mathcal{I}_Z\mathcal{F}_n = 0$ for all $n \geq 1$.*

Proof. We will use the criterion of Homology, Lemma 02MP. Moreover, we will use that if $0 \to (\mathcal{G}_n) \to (\mathcal{F}_n) \to (\mathcal{H}_n) \to 0$ is a short exact sequence of $Coh(X, \mathcal{I})$, then (a) $\mathcal{G}_n \to \mathcal{F}_n \to \mathcal{H}_n \to 0$ is exact for all $n \geq 1$ and (b) $\mathcal{G}_n$ is a quotient of $\mathrm{Ker}(\mathcal{F}_m \to \mathcal{H}_m)$ for some $m \geq n$. See proof of Lemma 087X.

Proof of (1). Let $(\mathcal{F}_n)$ be an object of $Coh(X, \mathcal{I})$. Then $\mathrm{Supp}(\mathcal{F}_n) = \mathrm{Supp}(\mathcal{F}_1)$ for all $n \geq 1$. Hence by remarks (a) and (b) above we see that for any short exact sequence $0 \to (\mathcal{G}_n) \to (\mathcal{F}_n) \to (\mathcal{H}_n) \to 0$ of $Coh(X, \mathcal{I})$ we have $\mathrm{Supp}(\mathcal{G}_1) \cup \mathrm{Supp}(\mathcal{H}_1) = \mathrm{Supp}(\mathcal{F}_1)$. This proves that the category defined in (1) is a Serre subcategory of $Coh(X, \mathcal{I})$.

Proof of (2). Here we argue the same way. Let $0 \to (\mathcal{G}_n) \to (\mathcal{F}_n) \to (\mathcal{H}_n) \to 0$ be a short exact sequence of $Coh(X, \mathcal{I})$. If $Z \subset X$ is a closed subscheme and $\mathcal{I}_Z$ annihilates $\mathcal{F}_n$ for all n, then $\mathcal{I}_Z$ annihilates $\mathcal{G}_n$ and $\mathcal{H}_n$ for all n by (a) and (b) above. Hence if $Z \to S$ is proper, then we conclude that the category defined in (2) is closed under taking sub and quotient objects inside of $Coh(X, \mathcal{I})$. Finally, suppose that $Z \subset X$ and $Y \subset X$ are closed subschemes proper over S such that $\mathcal{I}_Z \mathcal{G}_n = 0$ and $\mathcal{I}_Y \mathcal{H}_n = 0$ for all $n \geq 1$. Then it follows from (a) above that $\mathcal{I}_{Z \cup Y} = \mathcal{I}_Z \cdot \mathcal{I}_Y$ annihilates $\mathcal{F}_n$ for all n. By Lemma 3.6 (and via Definition 3.2 which tells us we may choose an arbitrary scheme structure used on the union) we see that $Z \cup Y \to S$ is proper and the proof is complete. □

4. Grothendieck's existence theorem, III

To state the general version of Grothendieck's existence theorem we introduce a bit more notation. Let A be a Noetherian ring complete with respect to an ideal I. Let $f : X \to \mathrm{Spec}(A)$ be a separated finite type morphism of schemes. Set $\mathcal{I} = I\mathcal{O}_X$. In this situation we let

$$Coh_{\text{support proper over } A}(\mathcal{O}_X)$$

be the full subcategory of $Coh(\mathcal{O}_X)$ consisting of those coherent $\mathcal{O}_X$-modules whose support is proper over $\mathrm{Spec}(A)$. This is a Serre subcategory of $Coh(\mathcal{O}_X)$, see Lemma 3.9. Similarly, we let

$$Coh_{\text{support proper over } A}(X, \mathcal{I})$$

be the full subcategory of $Coh(X, \mathcal{I})$ consisting of those objects $(\mathcal{F}_n)$ such that the support of $\mathcal{F}_1$ is proper over $\mathrm{Spec}(A)$. This is a Serre subcategory of $Coh(X, \mathcal{I})$ by Lemma 3.11 part (1). Since the support of a quotient module is contained in the support of the module, it follows that (0880) induces a functor

$$Coh_{\text{support proper over } A}(\mathcal{O}_X) \longrightarrow Coh_{\text{support proper over } A}(X, \mathcal{I}) \tag{4.0.1}$$

We are now ready to state the main theorem of this section.

Proof. We will use the equivalence of categories of Lemma 087T without further mention. For a closed subscheme $Z \subset X$ proper over A in this proof we will say a coherent module on X is "supported on Z" if it is annihilated by the ideal sheaf of Z or equivalently if it is the pushforward of a coherent module on Z. By Proposition 2.4 we know that the result is true for the functor between coherent modules and systems of coherent modules supported on Z. Hence it suffices to show that every object of $Coh_{\text{support proper over } A}(\mathcal{O}_X)$ and every object of $Coh_{\text{support proper over } A}(X, \mathcal{I})$ is supported on a closed subscheme $Z \subset X$ proper over A. This holds by definition for objects of $Coh_{\text{support proper over } A}(\mathcal{O}_X)$. We will prove this statement for objects of $Coh_{\text{support proper over } A}(X, \mathcal{I})$ using the method of proof of Proposition 2.4. We urge the reader to read that proof first.

Consider the collection Ξ of quasi-coherent sheaves of ideals $\mathcal{K} \subset \mathcal{O}_X$ such that the statement holds for every object $(\mathcal{F}_n)$ of $Coh_{\text{support proper over } A}(X, \mathcal{I})$ annihilated by $\mathcal{K}$. We want to show (0) is in Ξ. If not, then since X is Noetherian there

exists a maximal quasi-coherent sheaf of ideals $\mathcal{K}$ not in Ξ, see Lemma 01Y8. After replacing X by the closed subscheme of X corresponding to $\mathcal{K}$ we may assume that every nonzero $\mathcal{K}$ is in Ξ. Let $(\mathcal{F}_n)$ be an object of $Coh_{\text{support proper over } A}(X, \mathcal{I})$. We will show that this object is supported on a closed subscheme $Z \subset X$ proper over A, thereby completing the proof of the theorem.

Apply Chow's lemma (Lemma 0200) to find a proper surjective morphism $f : Y \to X$ which is an isomorphism over a dense open $U \subset X$ such that Y is H-quasi-projective over A. Choose an open immersion $j : Y \to Y'$ with Y' projective over A, see Morphisms, Lemma 01WA. Observe that

$$\text{Supp}(f^*\mathcal{F}_n) = f^{-1}\text{Supp}(\mathcal{F}_n) = f^{-1}\text{Supp}(\mathcal{F}_1)$$

The first equality by Morphisms, Lemma 056J. By assumption and Lemma 3.5 part (3) we see that $f^{-1}\text{Supp}(\mathcal{F}_1)$ is proper over A. Hence the image of $f^{-1}\text{Supp}(\mathcal{F}_1)$ under j is closed in Y' by Lemma 3.5 part (1). Thus $\mathcal{F}'_n = j_*f^*\mathcal{F}_n$ is coherent on Y' by Lemma 0CYJ. It follows that $(\mathcal{F}'_n)$ is an object of $Coh(Y', I\mathcal{O}_{Y'})$. By the projective case of Grothendieck's existence theorem (Lemma 1.3) there exists a coherent $\mathcal{O}_{Y'}$-module $\mathcal{F}'$ and an isomorphism $(\mathcal{F}')^\wedge \cong (\mathcal{F}'_n)$ in $Coh(Y', I\mathcal{O}_{Y'})$. Since $\mathcal{F}'/I\mathcal{F}' = \mathcal{F}'_1$ we see that

$$\text{Supp}(\mathcal{F}') \cap V(I\mathcal{O}_{Y'}) = \text{Supp}(\mathcal{F}'_1) = j(f^{-1}\text{Supp}(\mathcal{F}_1))$$

The structure morphism $p' : Y' \to \text{Spec}(A)$ is proper, hence $p'(\text{Supp}(\mathcal{F}') \setminus j(Y))$ is closed in $\text{Spec}(A)$. A nonempty closed subset of $\text{Spec}(A)$ contains a point of $V(I)$ as I is contained in the Jacobson radical of A by Algebra, Lemma 05GI. The displayed equation shows that $\text{Supp}(\mathcal{F}') \cap (p')^{-1}V(I) \subset j(Y)$ hence we conclude that $\text{Supp}(\mathcal{F}') \subset j(Y)$. Thus $\mathcal{F}'|_Y = j^*\mathcal{F}'$ is supported on a closed subscheme Z' of Y proper over A and $(\mathcal{F}'|_Y)^\wedge = (f^*\mathcal{F}_n)$.

Let $\mathcal{K}$ be the quasi-coherent sheaf of ideals cutting out the reduced complement $X \setminus U$. By Proposition 02O5 the $\mathcal{O}_X$-module $\mathcal{H} = f_*(\mathcal{F}'|_Y)$ is coherent and by Lemma 2.3 there exists a morphism $\alpha : (\mathcal{F}_n) \to \mathcal{H}^\wedge$ of $Coh(X, \mathcal{I})$ whose kernel and cokernel are annihilated by a power $\mathcal{K}^t$ of $\mathcal{K}$. We obtain an exact sequence

$$0 \to \text{Ker}(\alpha) \to (\mathcal{F}_n) \to \mathcal{H}^\wedge \to \text{Coker}(\alpha) \to 0$$

in $Coh(X, \mathcal{I})$. If $Z_0 \subset X$ is the scheme theoretic support of $\mathcal{H}$, then it is clear that $Z_0 \subset f(Z')$ set-theoretically. Hence Z_0 is proper over A by Lemma 3.3 and Lemma 3.5 part (2). Hence $\mathcal{H}^\wedge$ is in the subcategory defined in Lemma 3.11 part (2) and a fortiori in $Coh_{\text{support proper over } A}(X, \mathcal{I})$. We conclude that $\text{Ker}(\alpha)$ and $\text{Coker}(\alpha)$ are in $Coh_{\text{support proper over } A}(X, \mathcal{I})$ by Lemma 3.11 part (1). By induction hypothesis, more precisely because $\mathcal{K}^t$ is in Ξ, we see that $\text{Ker}(\alpha)$ and $\text{Coker}(\alpha)$ are in the subcategory defined in Lemma 3.11 part (2). Since this is a Serre subcategory by the lemma, we conclude that the same is true for $(\mathcal{F}_n)$ which is what we wanted to show. □

References

[1] Jean Dieudonné and Alexander Grothendieck. Éléments de géométrie algébrique. *Inst. Hautes Études Sci. Publ. Math.*, 4, 8, 11, 17, 20, 24, 28, 32, 1961–1967.